

University OF Engineering and Technology, Lahore
Computer Engineering Department

Course Name: PF	Course Code: CMPE-112L
Assignment Type: lab	Dated: 23 MARCH 2026
Semester: 1st	Session: Spring 2026
Lab/Project/Assignment #: 4	CLOs to be covered: CLO1
Lab Title: Nested Loop	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 4

Lab Goals/Objectives:

- **Traversing Multi-Dimensional Grids:** They allow you to easily iterate through rows and columns, such as processing 2D arrays, matrices, or pixel grids in image manipulation.
- **Generating Coordinate Pairings:** As seen in your tracking exercises, the outer loop sets a primary state (like an i -coordinate or row) while the inner loop completes a full cycle of secondary states (like a j -coordinate or column) for each step of the outer loop.
- **Handling Dynamic Geometric Patterns:** They enable the construction of complex text layouts (like half-pyramids and centered pyramids) by dynamically changing the inner loop's boundaries based on the outer loop's current iteration value.
- **Evaluating Combinations:** They are used to compare every element in a dataset with every other element, which is essential for algorithms like sorting (e.g., bubble sort) or finding pairs that meet specific conditions.

Equipment Required:

- Computer/Laptop
- Python 3.x
- VS Code / PyCharm / IDLE

Theory:

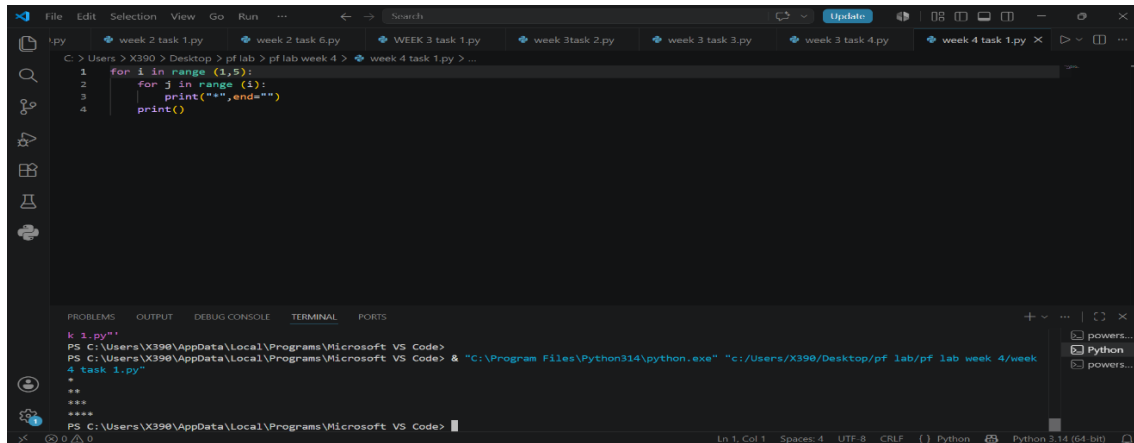
The theory of nested loops centers on placing a secondary loop entirely within the execution block of a primary master loop. For every single iteration of the outer loop, the inner loop initializes completely, executes its entire cycle, and terminates. This structural relationship establishes an independent control framework that natively maps out two-dimensional spaces, matrices, or coordinate matrices. The behavior can be static or dynamic, where modifying the inner range based on the outer tracking index alters the shape of the execution matrix over time. Ultimately, they serve as a fundamental algorithmic mechanism for evaluating multi-variable combinations, traversing grid data, and building repeating geometric patterns.

LAB TASKS:

TASK NO 1:

Write a Python program that utilizes nested `for` loops and the `range()` function to construct and display a right-angled, half-pyramid star (*) pattern on the console.

RESULT:



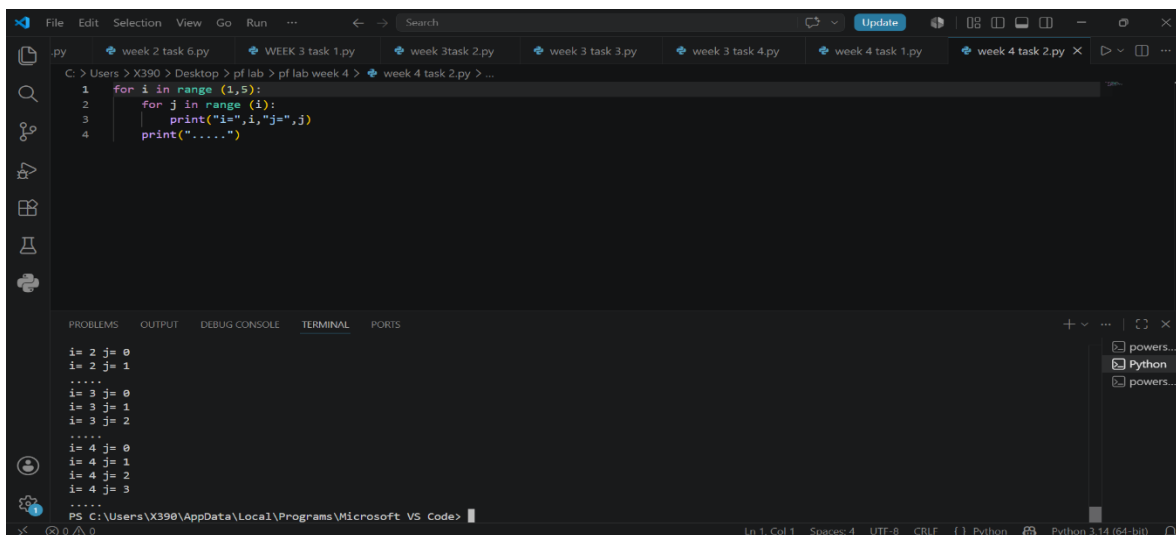
```
File Edit Selection View Go Run ... Search
week 2 task 1.py week 2 task 6.py WEEK 3 task 1.py week 3 task 2.py week 3 task 3.py week 3 task 4.py week 4 task 1.py
C:\Users\X390\Desktop> pf lab > pf lab week 4 > week 4 task 1.py > ...
1 for i in range(1,5):
2     for j in range(i):
3         print("*",end=" ")
4     print()

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
k 1.py"
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 4/week
4 task 1.py"
*
**
***
****
*****
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 2:

Write a Python program that utilizes nested `for` loops to systematically track, trace, and print the real-time index values of both an outer loop control variable and an inner loop control variable during execution. This task serves as a structural tracing exercise to understand how coordinate combinations change over time in nested environments

RESULT:



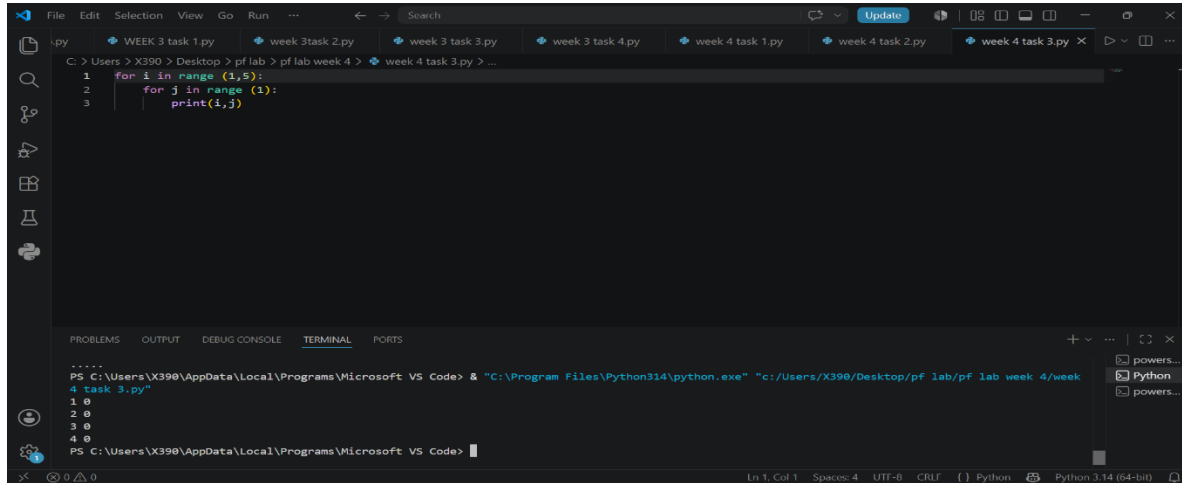
```
File Edit Selection View Go Run ... Search
week 2 task 6.py WEEK 3 task 1.py week 3 task 2.py week 3 task 3.py week 3 task 4.py week 4 task 1.py week 4 task 2.py
C:\Users\X390\Desktop> pf lab > pf lab week 4 > week 4 task 2.py > ...
1 for i in range(1,5):
2     for j in range(i):
3         print("i=",i,"j=",j)
4     print(".....")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
i= 2 j= 0
i= 2 j= 1
.....
i= 3 j= 0
i= 3 j= 1
i= 3 j= 2
.....
i= 4 j= 0
i= 4 j= 1
i= 4 j= 2
i= 4 j= 3
.....
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 3:

Write a Python program that utilizes nested `for` loops to demonstrate how a static inner loop threshold affects coordinate combinations. This task serves as a structural tracing exercise to contrast how a hardcoded inner loop boundary behaves compared to a dynamically changing inner loop boundary

RESULT:



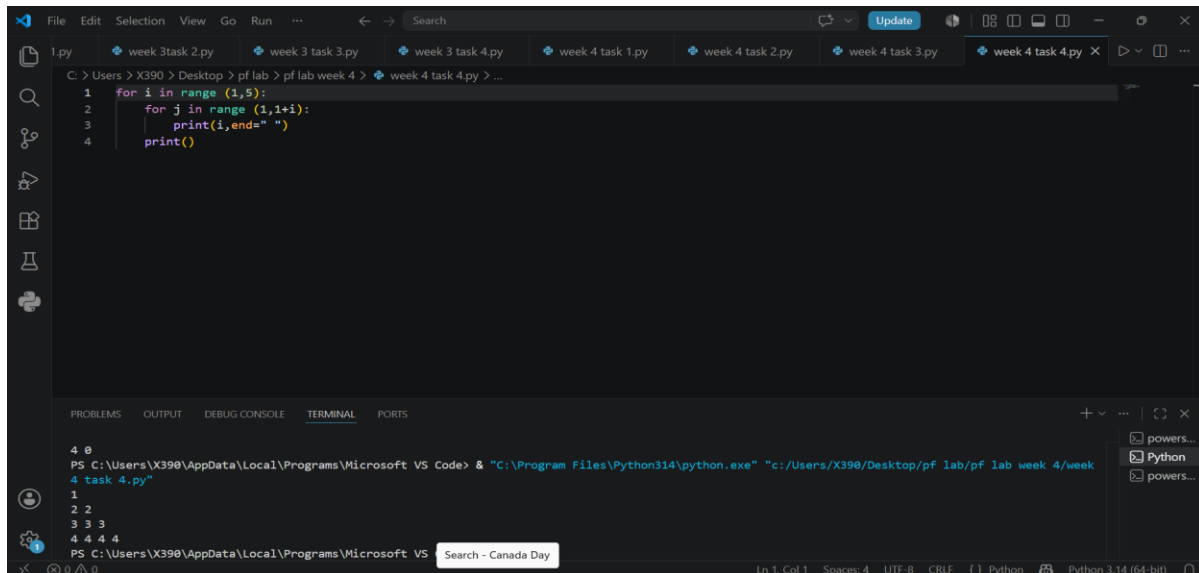
```
File Edit Selection View Go Run ... Search
1 for i in range(1,5):
2     for j in range(1):
3         print(i,j)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 4/week 4 task 3.py"
1 0
2 0
3 0
4 0
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 4:

Write a Python program that utilizes nested `for` loops and dynamic range structures to print a right-angled, half-pyramid number pattern on the console. Each line must display the current row number repeated a number of times equal to that row's index value.

RESULT:



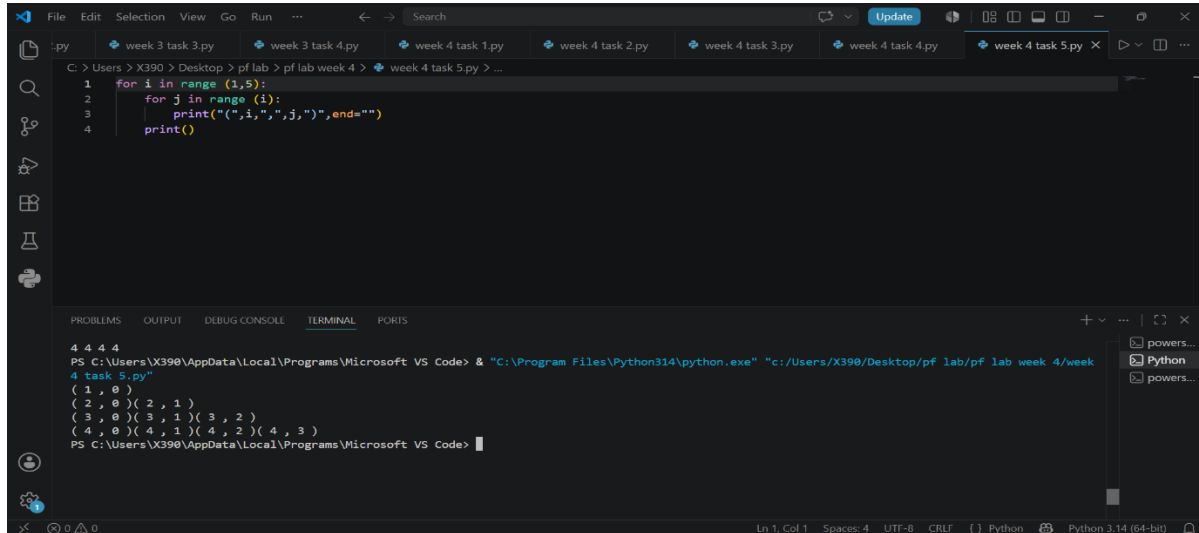
```
File Edit Selection View Go Run ... Search
1 for i in range(1,5):
2     for j in range(1,i+1):
3         print(i,end=" ")
4     print()

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 4/week 4 task 4.py"
1
2 2
3 3 3
4 4 4 4
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 5:

Write a Python program using nested `for` loops to construct and print a triangular matrix layout displaying 2D coordinate pairs (i, j) in real time. This task serves to reinforce understanding of dynamic inner loop boundaries and horizontal output formatting

RESULT:



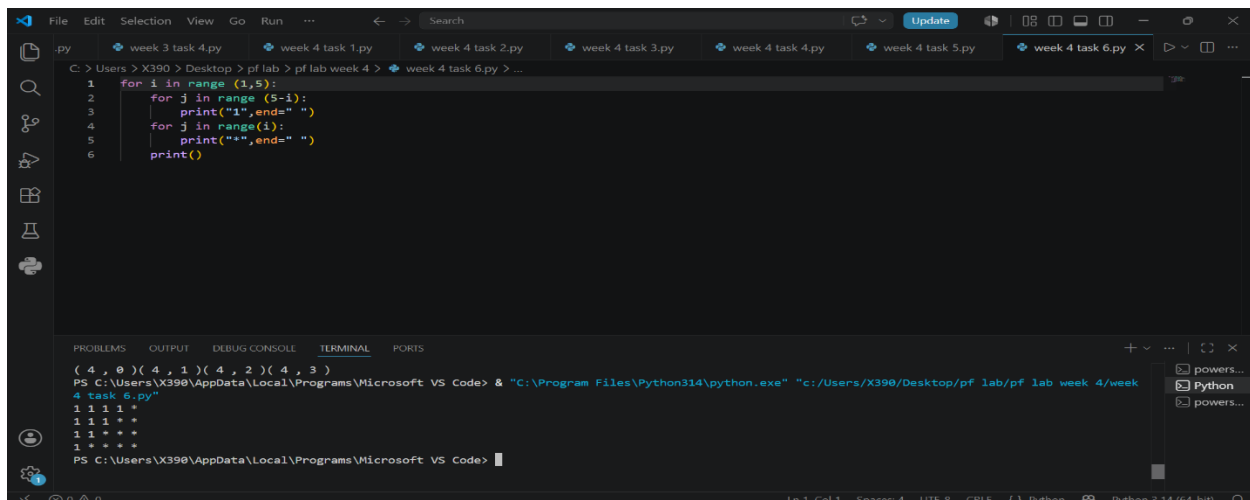
```
1 for i in range(1,5):
2     for j in range(i):
3         print("(" + i + "," + j + ")", end=" ")
4     print()
```

```
4 4 4 4
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 4/week 4 task 5.py"
( 1 , 0 )
( 2 , 0 ) ( 2 , 1 )
( 3 , 0 ) ( 3 , 1 ) ( 3 , 2 )
( 4 , 0 ) ( 4 , 1 ) ( 4 , 2 ) ( 4 , 3 )
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 6:

Write a Python program that utilizes multiple sequential inner loops nested within a single master outer loop to generate a 4x5 composite pattern grid. The program must print a linearly decreasing sequence of the numeric character "1" followed immediately on the same row by a linearly increasing sequence of asterisks (*).

RESULT:



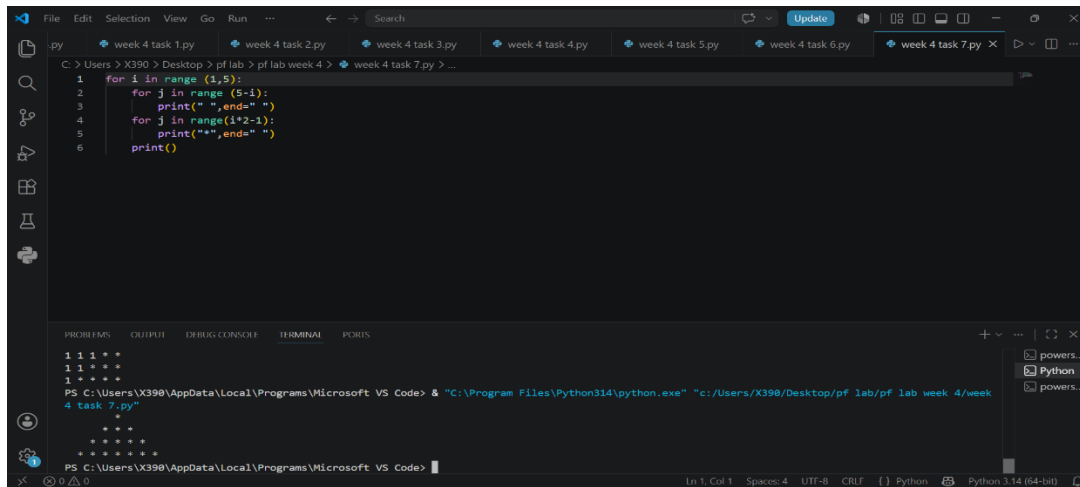
```
1 for i in range(1,5):
2     for j in range(5-i):
3         print("1", end=" ")
4     for j in range(i):
5         print("*", end=" ")
6     print()
```

```
( 4 , 0 ) ( 4 , 1 ) ( 4 , 2 ) ( 4 , 3 )
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 4/week 4 task 6.py"
1 1 1 1 *
1 1 1 *
1 1 *
1 *
```

TASK NO 7:

Write a Python program that utilizes multiple sequential inner loops nested within a single outer loop to generate a centered equilateral pyramid star (*) pattern on the console. The program must dynamically calculate and print a decreasing sequence of leading empty spaces followed immediately by an odd-numbered, increasing sequence of asterisks on each row

RESULT:



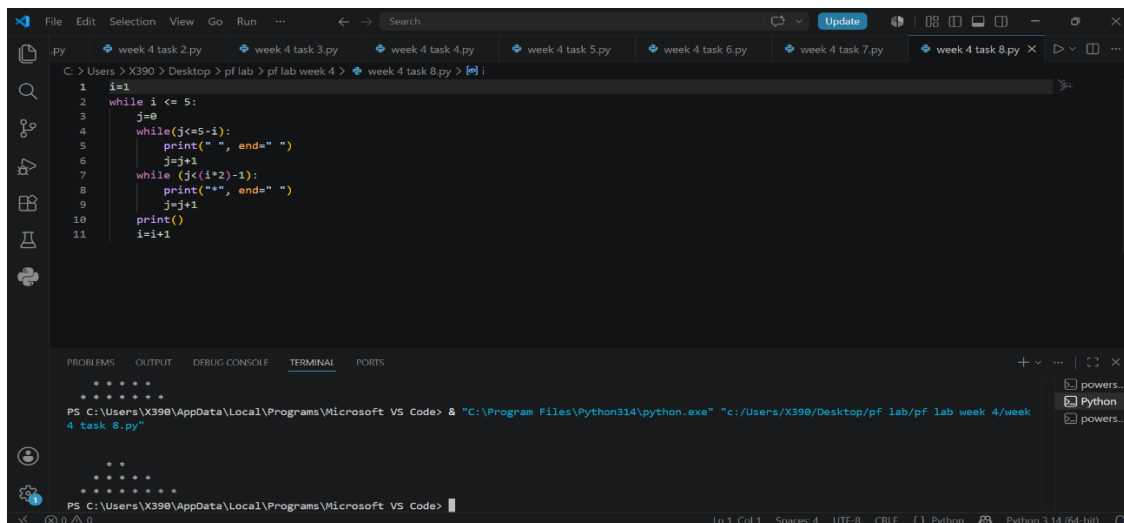
```
File Edit Selection View Go Run ... Search Update
.py week 4 task 1.py week 4 task 2.py week 4 task 3.py week 4 task 4.py week 4 task 5.py week 4 task 6.py week 4 task 7.py x
C:\Users\X390\Desktop> pf lab > pf lab week 4 > week 4 task 7.py > ...
1 for i in range(1,5):
2     for j in range(5-i):
3         print(" ",end=" ")
4     for j in range(i*2-1):
5         print("*",end=" ")
6     print()

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
1 1 1 * *
1 1 * * *
1 * * * *
* * * * *
* * * * *
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 4/week
4 task 7.py"
*
* *
* * *
* * * *
* * * * *
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 8:

Write a Python program that replicates a centered pyramid star pattern using indefinite `while` loops instead of definite `for` loops. The script must manually control and maintain state variables for row selection, leading indentation spacing, and odd-numbered star distribution.

RESULT:



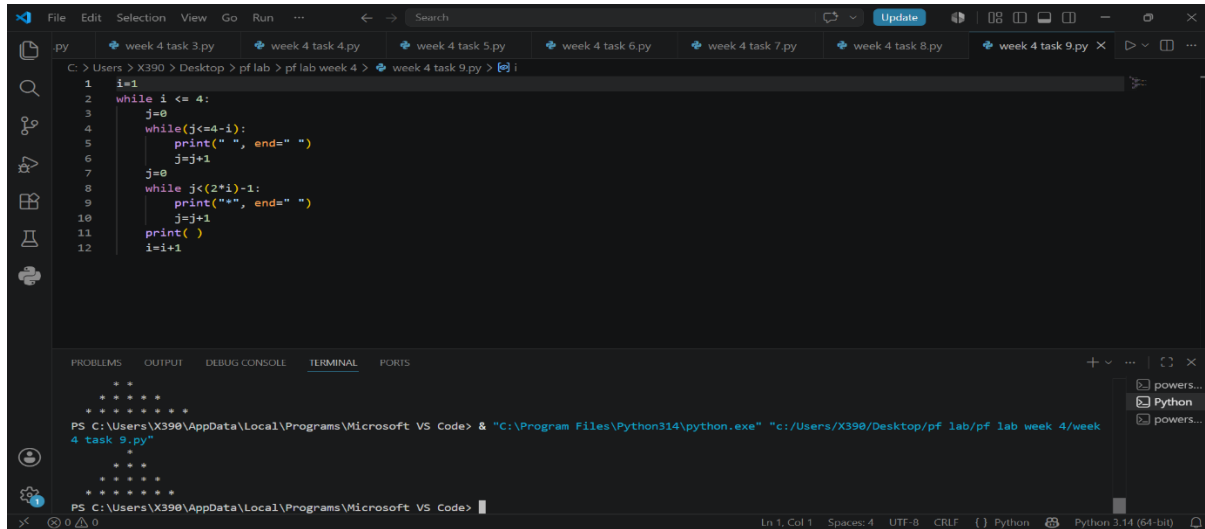
```
File Edit Selection View Go Run ... Search Update
.py week 4 task 2.py week 4 task 3.py week 4 task 4.py week 4 task 5.py week 4 task 6.py week 4 task 7.py week 4 task 8.py x
C:\Users\X390\Desktop> pf lab > pf lab week 4 > week 4 task 8.py > ...
1 i=1
2 while i <= 5:
3     j=0
4     while(j<=5-i):
5         print(" ",end=" ")
6         j=j+1
7     while (j<(i*2)-1):
8         print("*",end=" ")
9         j=j+1
10    print()
11    i=i+1

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> & "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 4/week
4 task 8.py"
*
* *
* * *
* * * *
* * * * *
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code>
```

TASK NO 9:

Write a Python program using indefinite `while` loops to generate a perfectly centered, equilateral pyramid star (*) pattern on the console. This assignment serves to fix the state-tracking variable bugs from previous iterations by implementing proper variable resets prior to nested conditional blocks.

RESULT:



```
1 i=1
2 while i <= 4:
3     j=0
4     while(j<=4-i):
5         print(" ", end=" ")
6         j=j+1
7     j=0
8     while j<(2*i)-1:
9         print("*", end=" ")
10        j=j+1
11    print()
12    i=i+1
```

```
PS C:\Users\X390\AppData\Local\Programs\Microsoft VS Code> "C:\Program Files\Python314\python.exe" "c:/Users/X390/Desktop/pf lab/pf lab week 4/week 4 task 9.py"
  *
 * *
* * *
* * * *
```

CONCLUSION:

A nested loop—a loop placed inside the body of another loop—is a powerful programming construct essential for working with multi-dimensional data structures and complex repetitive patterns. Its core mechanism relies on a multiplicative relationship: for every single iteration of the outer loop, the inner loop executes its entire cycle from start to finish.

While nested loops are indispensable for tasks like traversing grids (matrices), comparing every item in a list to every other item, or rendering multi-layered graphics, they come with a significant performance trade-off